# 11.8 Tilted Equilibria

WORKSHEET

**The question**

**STEP III 2025, Question 9**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

**Ask yourself** A solid rests on a table, touching at a single point $X$ of its curved surface. The forces on it are its weight (acting at the centre of mass $G$) and the reaction at $X$. When are these in equilibrium?

### STAGE 1 The centre of mass

**Ask yourself** Slice the solid into thin cylindrical shells of radius $x$, thickness $\delta x$ and height $y$. What is the volume of a shell, and where is its centre of mass?

### STAGE 2 Where normals meet the axis

**Ask yourself** For $n = 4$, $Y = r(1 - p^4 - \frac{1}{4p^2})$. Where is it greatest? What happens as $p \to 0$, and at $p = 1$?

**STAGE 3** **The spinning top**

**Ask yourself** The solid in part (iii) is a dome ($n = 4$) on top of an upside-down paraboloid ($n = 2$). Where is its centre of mass?

**STAGE 4** **Counting equilibria**

**Ask yourself** A tilted equilibrium needs a point (not $A$ or $B$) whose normal passes through $G$. On each part of the surface, how many points have normals meeting the axis at height $\frac{3r}{35}$?

## What's really going on

### Equilibria as critical points

When the solid rests on the point $X$, the height of $G$ above the table is the distance from $G$ to the tangent plane at $X$, which for a convex body is $|GX|$ when $GX$ is along the normal. In general, the equilibria are the *critical points* of the distance from $G$ to the surface: a point $X$ where $|GX|$ is stationary is exactly a point whose normal passes through $G$. Stable equilibria are local minima of this distance ($G$ as low as possible), and unstable ones are local maxima or saddle points.

Going round the cross-section from $A$ to $B$, the distance from $G$ is: large at $A$ (a maximum: balanced on its point, unstable), smaller at the tilted point on the base (a minimum, stable), larger at the rim, smaller at the lower tilted point of the dome (stable), larger at the upper one (unstable), and smaller at $B$ (stable: resting on the top of the dome). Minima and maxima must alternate, as the figure shows. (In fact there is one more equilibrium that the question does not count: the solid can balance, unstably,

on its sharp rim, where the surface has no single normal.)

#### The Gömböc

How few equilibria can a convex body have? Any body has at least one stable and one unstable position: the nearest and furthest points of the surface from $G$. A two-dimensional convex shape of uniform density always has at least four (two minima and two maxima of distance, alternating). In 1995 Vladimir Arnold asked whether a three-dimensional convex body of uniform density could have just two: one stable, one unstable. In 2006 Gábor Domokos and Péter Várkonyi constructed one, the *Gömböc*: put it down in any position, and it rolls and rocks until it reaches its single stable position. Domokos later noticed that some tortoises have shells close to this shape, which helps them right themselves when turned over.

#### Archimedes and floating bodies

The study of the stability of bodies goes back to Archimedes, who in *On Floating Bodies* analysed when a floating paraboloid (a model of a ship's hull) is stable upright and when it capsizes, in terms of its shape and density. His answer, worked out with geometry alone, involves exactly the kind of centre-of-mass and normal calculations used here. Naval architects still use his ideas, in the form of the *metacentre*: a ship is stable if the point where the vertical through the centre of buoyancy meets the ship's axis lies above the centre of mass.

## Try this next

1 **A self-righting toy.** Try STEP II 1996, Question 9 (step.maths.org/questions/96-s2-q9): when does a cone on a hemisphere right itself? (*Hint:* where must $G$ be relative to the centre of the hemisphere?)

2 **A sphere.** For a uniform solid sphere, every normal passes through the centre. What does that say about its equilibria? What if the sphere's centre of mass is displaced slightly from its centre?

3 **An ellipse.** Show that an elliptical disc rolling in its own plane on a line has exactly four equilibria, and identify which are stable.

4 **Two paraboloids.** Replace the dome in part (iii) by a paraboloid $ry = r^2 - x^2$, so that the solid is symmetric about the $x$-axis. Show that $G$ is at the origin, find the tilted equilibria, and explain why they come in pairs with equal tilts.

## Further reading

► **Péter Várkonyi & Gábor Domokos, "Static equilibria of rigid bodies: dice, pebbles, and the Poincaré–Hopf theorem"**, *Journal of Nonlinear Science* 16 (2006). The Gömböc paper.

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), on equilibrium and stability.
- **Chris Rorres, "Completing Book II of Archimedes's *On Floating Bodies*"**, *Mathematical Intelligencer* 26 (2004). Archimedes' stability analysis, with modern pictures.


The full guided solution is in Section 11.8:
https://beyondstep.keithmcnulty.org/ch11/tilted-equilibria.html